Estimating True Movie Quality and Temporal Rating Bias with a Bayesian Hierarchical Model

Bayesian Analysis

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Abstract

This paper presents a progressive hierarchical Bayesian framework designed to analyze recommendation engines while decomposing platform trends, user behavior, and catalog polarization. Using MCMC estimation in Stan, Model A isolates global temporal drift via non-linear B-splines to establish an unbiased baseline. Model B introduces personalization through an individual user scale shift and a high-dimensional user-by-genre preference matrix, successfully decoupling a reviewer's leniency from their specific genre tastes to reduce unexplained user variance. Finally, Model C accommodates catalog polarization by introducing movie specific variance. Integrated into a full-chain Posterior Predictive Distribution engine with a genre filter, Model C maps the risk-return by calculating expected ratings alongside explicit downside failure probabilities.

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1 Introduction.

1.1 Problem Description

When you look at a movie streaming platform today, almost everything relies on a single number: the average star rating. If a movie has a 4.0-star average, the system treats it as a good recommendation. But this simple approach hides a massive flaw. A 4.0-star average can mean completely different things. Therefore, building a reliable movie recommendation system is deceptively difficult because the raw user rating data are fundamentally contaminated by systematic behavioral noise. When a platform bases its rankings or recommendations on simple empirical star averages, it falls into critical traps caused by four distinct types of data bias:

1. Temporal Bias (Nostalgia Effect)
2. Law of Small Numbers
3. User Harshness
4. Audience Polarization

Failing to isolate and correct these biases means that recommendations remain highly inaccurate, flattening out real preferences, and exposing users to high-risk, frustrating viewing experiences.

1.2 Objectives

The ultimate objective of this project is to filter out these compounding biases and uncover the true latent quality of each film. The scope of the work involves constructing a progressive sequence of Bayesian hierarchical frameworks designed to systematically neutralize noise. Specifically, the project aims to:

1. Neutralize temporal drift (nostalgia bias) and individual user harshness scales simultaneously
2. Implement robust Bayesian shrinkage to pull unstable, low-volume movie scores toward the global mean, neutralizing the law of small numbers.
3. Quantify audience polarization directly by treating movie variance as an active parameter space, allowing the engine to separate broad consensus appeal from divisive cult titles.
4. Create a recommendation system based on all these parameters for each user.

1.3 Response and Explanatory Variables

To capture these real-world dynamics mathematically, our statistical framework defines the following structural variables:

- Response Variable (y_{ij}): The observed numerical rating given by user j to movie i, bounded on a continuous 1.0 to 5.0 star scale.
- Explanatory Variables:
  - Movie Index (i): Unique identifier used to isolate the true underlying quality and capture the parameters of disagreement of the audience specific to the item.
  - User index (j): Unique identifier mapping specific user baselines to measure individual grading severity or indulgence
  - Movie Age (age_i): The time elapsed since the movie's theatrical release until user i submitted his review, used to capture and neutralize the temporal nostalgia curve.
  - Genre (k): The specific genre designation of a film, which acts as a key component to track personal genre attraction coefficients

1.4 Data Presentation

To see how this works in practice, we train our models on a raw slice of historical movie ratings that contains all the structural pieces needed to clear away user and temporal biases. Instead of looking at movie scores as isolated numbers, our dataset tracks the specific ingredients behind every single interaction. By recording the User ID, we can identify whether a review came from a naturally harsh critic or an easy grader and adjust for their personal baseline. Tracking the Movie ID lets us see exactly how many reviews a film has, allowing the system to ignore shaky scores from obscure titles with low sample sizes. Measuring Movie Age gives us a timeline to map out and erase the "nostalgia effect" that artificially boosts older films.

userId	movieId	title	genres	movie_age	user_idx	movie_idx	genre_idx	rating (y_{ij})
1	1	Toy Story (1995)	Adventure|Animation|Children|Comedy|Fantasy	5	1	1	352	4.0
1	6	Heat (1995)	Action|Crime|Thriller	5	1	6	261	4.0
1	47	Seven (1995)	Mystery|Thriller	5	1	44	938	5.0
1	70	From Dusk Till Dawn (1996)	Action|Comedy|Horror|Thriller	4	1	63	215	3.0
1	110	Braveheart (1995)	Action|Drama|War	5	1	98	291	4.0
1	223	Clerks (1994)	Comedy	6	1	191	635	3.0

Table 1: Representative Sample Slice of Raw Input Data Matrix

A major hurdle discovered during the initial inspection of the dataset involves how film genres are recorded. In the raw data, genres are not grouped cleanly; instead, they appear as long strings of piped combinations, creating nearly 1,000 unique, highly specific genre strings across the catalog. Attempting to build an individual model parameter for 1,000 sparse groups would lead to severe overfitting and computational collapse. To make the data mathematically workable while still preserving the unique theme of each film, these raw strings were grouped together into eight major foundational genre categories. To see how this consolidation looks after processing, a sample slice of the mapped dataset—including the newly assigned group index keys is presented below:

userId	movieId	rating (y_{ij})	title	genres	genre_group	user_idx	movie_idx	genre_idx
1	1	4.0	Toy Story (1995)	Adventure|Animation|Children|Comedy|Fantasy	Action	1	1	1
1	3	4.0	Grumpier Old Men (1995)	Comedy|Romance	Romance	1	3	7
1	6	4.0	Heat (1995)	Action|Crime|Thriller	Action	1	6	1
1	47	5.0	Seven (a.k.a. Se7en) (1995)	Mystery|Thriller	Horror_Thriller	1	44	5
1	50	5.0	Usual Suspects, The (1995)	Crime|Mystery|Thriller	Horror_Thriller	1	47	5
1	70	3.0	From Dusk Till Dawn (1996)	Action|Comedy|Horror|Thriller	Action	1	63	1
1	101	5.0	Bottle Rocket (1996)	Adventure|Comedy|Crime|Romance	Action	1	90	1
1	110	4.0	Braveheart (1995)	Action|Drama|War	Action	1	98	1

Table 2: Processed Movie Ratings Slice with Simplified Genre Categories and Matrix Index Keys

2 Analysis

2.1 Exploratory Data Analysis (EDA)

Before building our predictive bayesian frameworks, we must look closely at the raw data to uncover the systematic noise that corrupts regular movie ratings. Our descriptive analysis reveals the major behavioral patterns mentioned int he problem descriptive section that prove why simple average scores fail.

The Nostalgia bias

The first structural issue we identify is related to time. When we plot the average movie rating against the age of a movie in decades, we find clear visual evidence of a Nostalgia Effect. Rather than staying flat, the average rating line steadily climbs as movies get older, peaking noticeably for films released between 30 and 60 years ago. This upward curve proves that older movies carry an artificial rating advantage. If we do not account for this temporal drift, newer high-quality releases are unfairly penalized when compared directly against legacy classics that benefit from historical nostalgia.

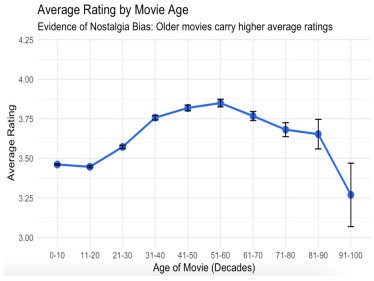


Figure 1: Average Rating vs Movie Age

In addition, we can see as movies get older than 70 years the standard error increases proving there are not a large number of reviews for the movies with this age or older which takes us to the second bias.

The law of Small Numbers

The second major hurdle is extreme data sparsity, which is exposed beautifully by our Funnel Plot. In this chart, we map each movie's average rating against its total number of reviews. The plot creates a distinct funnel shape: for movies with hundreds of reviews, the data points tightly cluster above the true global platform average which makes sense as people tend to watch movies that many people agrees on its quality. However, as we look to the left side of the chart where movies have very low sample sizes, the dots violently scatter to the absolute limits of the scale, showing "fake" extreme averages of perfect 5.0s or terrible 1.0s. This proves that obscure, low-volume movies naturally dominate the top and bottom of unadjusted lists due to sheer mathematical luck, justifying the need for Bayesian shrinkage to pull these unstable scores back toward the global center.

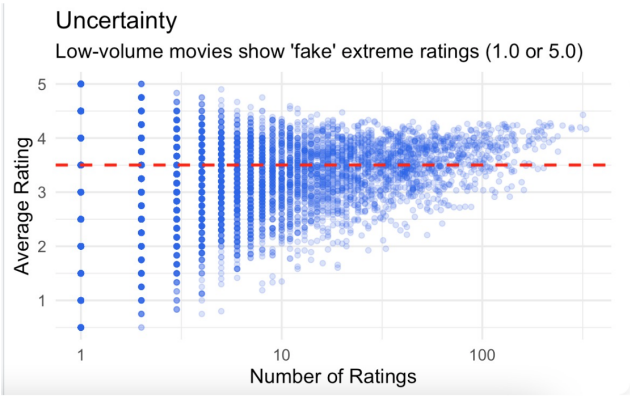


Figure 2: Funnel Plot

User Harshness

The final piece of noise relates to the judges themselves. When we plot a histogram showing the number of users against the average rating that each user gives, we get a wide distribution that proves people do not use the 1-to-5 star scale uniformly. The data naturally splits our audience into two distinct camps: The Grumps and The Optimists. "Grumps" sit on the left tail of the histogram, meaning their internal grading baseline is consistently 1 to 2 stars below the platform average. Conversely, "Optimists" cluster on the far right, using the upper end of the scale exclusively. This spread proves that a raw rating cannot be trusted on its own; a 3-star rating from a severe "Grump" actually signals a brilliant film, while a 4-star rating from an "Optimist" might just mean the movie was mediocre.

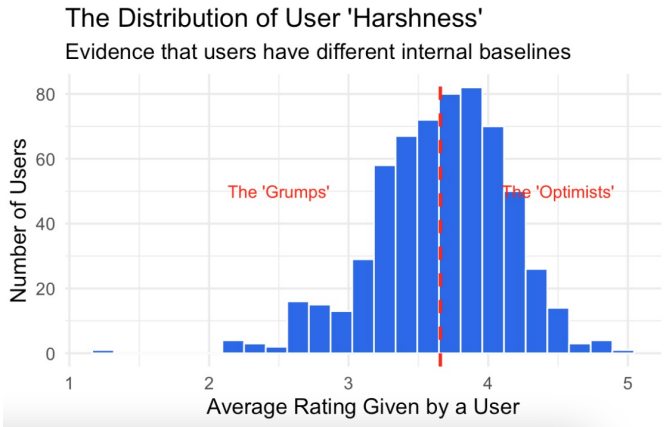


Figure 3: User Harshness

2.2 Bayesian Hierarichal Analysis

To systematically isolate and remove the behavioral noise patterns identified in our descriptive analysis, we constructed three progressive Bayesian hierarchical frameworks. Each configuration builds upon the last by expanding the parameter space to capture increasingly complex relationships within our user interaction data matrix.

B-Splines for Nostalgia Adjustment

Our Exploratory Data Analysis clearly demonstrated that the Nostalgia Bias is non-linear; movie ratings don't just constantly increase or decrease with age, but rather follow a distinct curve that peaks for films released 30 to 60 years ago. To confirm this assumption objectively before modeling, a residual analysis was conducted.

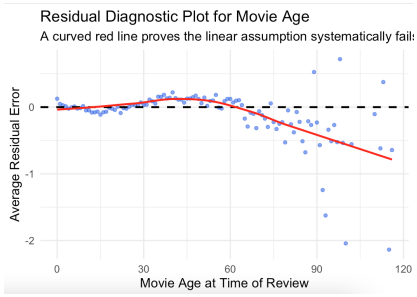


Figure 4: Residual Analysis

A standard linear model's residuals were plotted against movie age. As shown in the plot above, the resulting fit shows a distinct curved red line rather than a random cloud of points around 0. This distinct pattern in the residuals proves that a simple linear model fails to capture the temporal noise, leaving behind a systematic bias that must be addressed.

B-splines function by chopping the entire timeline of movie history into distinct chronological segments divided by boundaries called knots. Within each segment, a localized polynomial curve is fitted to the data. Where the segments meet at the knots, they are mathematically forced to connect seamlessly and smoothly. Instead of a single movie age column, the variable is transformed into a set of basis functions (B_k (age)). When our Bayesian models run, they estimate an individual coefficient (γ_k) for each piece of the spline. When these pieces are summed together, they construct a perfectly fluid, highly accurate non-linear curve that maps out the exact footprint of the Nostalgia Effect.

For any given movie age, its representation is expanded across K basis functions:

$$\text{Nostalgia Curve(age}i) = \sum_{k=1}^{K} \gamma_k B_{i,k} \tag{1}$$

where

- $B_{i,k}$: is the value of the k-th basis spline evaluated at $movie_i$ specific age.
- γ_k : is the weight parameter estimated by the model for that specific time interval.

To sum up this section, rather than treating time as a single straight line, a B-spline works like a flexible, connecting track that chops history into distinct chronological segments. For any given film, the spline determines exactly how its age distributes across these neighboring eras, assigning it a set of proportional weights. When the Bayesian model runs, it calculates a specific global "nostalgia penalty or bonus" for each era across millions of reviews. By multiplying the movie's era proportions by these calculated weights, the system constructs a perfectly fluid, continuous curve that maps the non-linear footprint of time bias. This allows the model to cleanly isolate and subtract out the artificial score inflation a film receives simply for being old, leaving behind its true, uncorrupted latent quality.

2.2.1 Model A: Baseline Filter

Model A established the foundation of our bayesian hierarchical framework. Its main goal is to balance the true quality of each movie against Nostalgia Bia and determine whether an individual rating came from a harsh or lenient critic.

$$Y_{i,j} \sim \text{Normal}(\mu_{i,j}, \sigma^2) \tag{2}$$

$$\mu_{i,j} = \beta_0 + \alpha_i + \delta_j + \sum_{k=1}^{K} \gamma_k B_{i,k} \tag{3}$$

Parameter Specifications

Parameter	Description	Prior Distribution
β_0	**Global Average Rating:** The baseline intercept across all items and users on the streaming platform.	Normal$(3.5, 1)$
α_i	**Movie Quality Deviation:** The isolated deviation for movie i from the global average. It captures true structural film quality.	Normal$(0, \sigma_i^2)$
δ_j	**Review Harshness Deviation:** The adjustment factor for user j. It shifts negative for "Grumps" (severe critics) and positive for "Optimists" (lenient reviewers).	Normal$(0, \sigma_j^2)$
γ_k	**Spline Weight Vector:** The continuous adjustment weights applied to the B-spline basis vectors to isolate and scrub out the non-linear temporal bias.	Normal$(0, 1)$

Table 3: Model Parameter Specifications and Prior Distributions

2.2.2 Model B: Genre Attraction Filter

While Model A successfully balances overall platform biases, it operates under the strict assumption that individual tastes are completely uniform across different styles of film. In reality, a user's review profile is heavily influenced by their personal affinity for certain genres. Model B expands upon our baseline framework by adding a user-specific taste interaction parameter, $\theta_{j,k}$. The core objective of Model B is to establish a "genre-neutral" baseline for true movie quality (α_i). Without isolating this taste interaction, an excellent Sci-Fi or Horror movie might be heavily penalized in its quality ranking simply because it happened to be reviewed by a cluster of users who naturally dislike those genres. By pulling out these personalized variations, the model can separate a user's systematic critique tendencies ("Grumps" vs. "Optimists") from their underlying thematic biases.

$$Y_{i,j} \sim \text{Normal}(\mu_{i,j}, \sigma^2) \tag{4}$$

$$\mu_{i,j} = \beta_0 + \alpha_i + \delta_j + \theta_{j,k} + \sum_{k=1}^{K} \gamma_k B_{i,k} \tag{5}$$

Parameters Specifications

Parameter	Description	Prior Distribution
β_0	**Global Average Rating:** The baseline intercept across all items and users on the streaming platform.	Normal$(3.5, 1)$
α_i	**Movie Quality Deviation:** The isolated deviation for movie i from the global average, capturing true structural film quality.	Normal$(0, \sigma_i^2)$
δ_j	**Review Harshness Deviation:** The adjustment factor for user j. Shifts negative for "Grumps" (severe critics) and positive for "Optimists" (lenient reviewers).	Normal$(0, \sigma_j^2)$
$\theta_{j,k}$	**Personal Genre Affinity:** Measures how much user j naturally likes or dislikes genre category k, pulling out taste from the structural movie quality baseline.	Uniform$(-1.5, 1.5)$
γ_k	**Spline Weight Vector:** The continuous adjustment weights applied to the B-spline basis vectors to isolate and scrub out the non-linear temporal bias.	Normal$(0, 1)$
$\sigma_i, \sigma_j, \sigma$	**Variance Hyperparameters:** Scale parameters governing the global observation error and the group-level shrinkage vectors.	Exponential(1)

Table 4: Model B Parameter Specifications, Functional Descriptions, and Prior Distributions

2.2.3 Model C

The previous configurations assume homoskedasticity meaning they assume every movie shares the exact same error variance (σ), regardless of how controversial it might be. Model C eliminates this flaw by introducing heteroskedasticity, acknowledging that some movies are fundamentally divisive cult films, while others are broad appeal consensus entries.

By turning the variance scale into an item-specific parameter space (σ_i) Model C can capture audience disagreement. A small (σ_i) means the movie has uniform consensus across different users, making its predicted rating highly reliable.

A large (σ_i) flags a highly volatile "cult classic" where users deeply disagree on its quality. This item-specific variance parameter provides a valuable risk metric for recommendation engines, allowing the system to distinguish between a "safe" 4.0-star movie and a "risky" 4.0-star movie.

$$Y_{i,j} \sim \text{Normal}(\mu_{i,j}, \sigma_i) \tag{6}$$

$$\mu_{i,j} = \beta_0 + \alpha_i + \delta_j + \theta_{j,k} + \sum_{k=1}^{K} \gamma_k B_{i,k} \tag{7}$$

The uniform global variance parameter is replaced with an item-specific variance term, (σ_i) making the scale an active parameter space. To guarantee that the noise scale remains strictly positive during Bayesian estimation, we define the parameter transformation in log-space:

$$\log(\sigma_i) = \sigma_{\text{global}} + \eta_i \tag{8}$$

Parameters Specifications

Parameter	Description	Prior Distribution
β_0	**Global Average Rating:** The baseline intercept across all items and users on the streaming platform.	Normal$(3.5, 1)$
α_i	**Movie Quality Deviation:** The isolated deviation for movie i from the global average, capturing true structural film quality.	Normal$(0, \sigma_i^2)$
δ_j	**Review Harshness Deviation:** The adjustment factor for user j. Shifts negative for "Grumps" (severe critics) and positive for "Optimists" (lenient reviewers).	Normal$(0, \sigma_j^2)$
$\theta_{j,k}$	**Personal Genre Affinity:** Measures how much user j naturally likes or dislikes genre category k, pulling out idiosyncratic taste from the structural movie quality baseline.	Uniform$(-1.5, 1.5)$
γ_k	**Spline Weight Vector:** The continuous adjustment weights applied to the B-spline basis vectors to isolate and scrub out the non-linear temporal bias.	Normal$(0, 1)$
σ_{global}	**Global Noise Floor:** The log-space baseline variability for the standard deviation across the entire movie catalog.	Normal$(0.5, 0.5)$
η_i	**Audience Disagreement Factor:** The item-specific polarization metric. Positive values indicate highly controversial cult films; negative values reveal uniform consensus titles.	Uniform$(-1.0, 0.5)$
σ_i, σ_j	**Variance Hyperparameters:** Scale parameters governing the group-level shrinkage vectors for movies and users.	Exponential(1)

Table 5: Model C Parameter Specifications, Functional Descriptions, and Prior Distributions

2.3 Results

Bayesian Shrinkage

Before analyzing the unique predictive performance of each model, we examine the global parameters and structural adjustments that remain consistent across our entire hierarchical framework. Accounting for these global parameters standardizes the scoring matrix, allowing us to isolate personal tastes and true movie quality from widespread catalog biases.

One of the objectives mentioned at the beginning of the report was to implement robust Bayesian shrinkage to solve the law of small number bias by moving the scores toward the global mean.

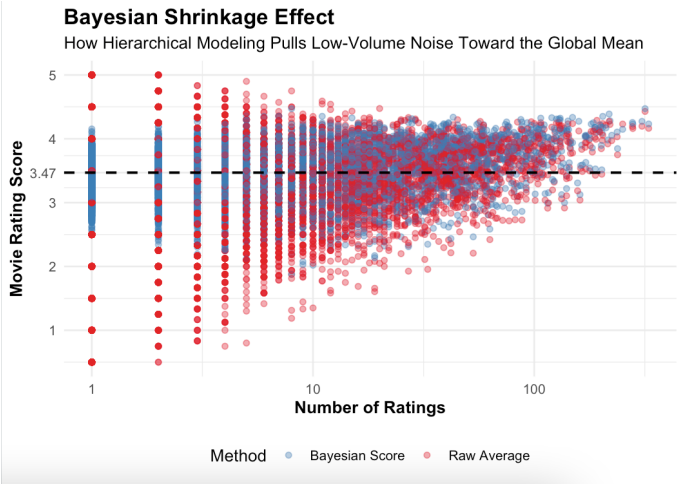


Figure 5: Bayesian Funnel Plot

As shown in the funnel plot above Across all three models, the global intercept (β_0) converged cleanly around a posterior mean of 3.47 represented by the horizontal dashed line. The group-level hyperparameters demonstrate the powerful effect of Bayesian shrinkage when contrasting raw data calculations against our hierarchical estimations.

As proven in the Bayesian Shrinkage funnel plot, there is a clear behavioral divide based on sample size:

- Low-Volume Data Left Tail: In this sparse data region, the red dots (Raw Averages) violently scatter across the entire vertical axis, showing highly unstable, "fake" extreme scores. Conversely, the blue dots (Bayesian Scores) show our hierarchical model actively pulling these low-volume records tightly back toward the global platform mean of 3.47. The model correctly recognizes that these extreme scores are driven by small sample size luck rather than true quality, heavily compressing their variance
- High-Volume Data Right Tail: As the number of ratings increases along the x-axis, the data volume naturally provides high statistical certainty. Here, the red raw averages and the blue Bayesian scores perfectly converge and blend together. Crucially, the plot reveals an upward trend in this region, demonstrating that movie rating scores naturally skew higher as volume increases. This reflects films that accumulate hundreds or thousands of ratings are typically broad-appeal favorites or critically acclaimed hits. The Bayesian framework adapts seamlessly here: it aggressively shrinks scores when data is scarce, but steps aside to let the data speak for itself once a title proves its popularity through high review volume.

Posterior Distribution of the Nostalgia B-Splines

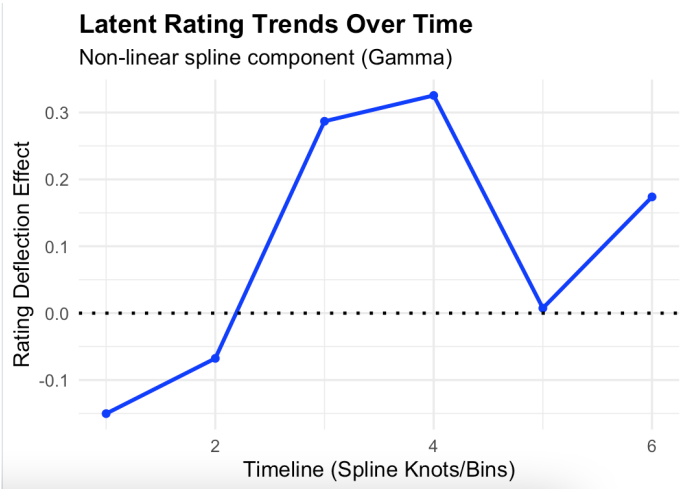


Figure 6: Bayesian Funnel Plot

The posterior estimates for the B-spline coefficients (γ_k) plotted across the timeline knots confirm that temporal noise is highly non-linear, validating the decision to use a flexible framework rather than a rigid linear model. By plotting the platform deflection effects against the sequential spline bins, the model uncovers distinct patterns:

- The Modern Era Penalty (Bins 1–2): For the newest titles on the platform, the spline coefficients sit firmly in negative territory, starting at a deflection of approximately -0.15 at Bin 1 and rising slightly to -0.07 at Bin 2.

This statistically confirms a "recency penalty," where users tend to grade contemporary or newly released content more harshly compared to the historical baseline.

- The Nostalgia Peak (Bins 3–4): As we move further back into the timeline, the curve aggressively swings upward into positive territory. The rating deflection peaks sharply between Bin 3 (+0.29) and Bin 4 (+0.32). This substantial upward jump provides a robust evidence of an existing nostalgia premium, where movies from these specific ages receive a significant scoring boost simply due to historical sentiment and platform bias.
- Time Neutralization(Bins 5–6): Following the nostalgia peak, the bias drops sharply back down to near zero at Bin 5 (+0.01), showing an era where time-based sentiment exerts almost no influence on the ratings. However, moving into the oldest vintage category at Bin 6, the curve rebounds back up to +0.17, capturing a secondary wave of classic appreciation.

2.3.1 Model A

To estimate the posterior distributions, Markov Chain Monte Carlo (MCMC) sampling was executed using 2 parallel chains for 1,000 iterations per chain across all models. The empirical summary metrics for Model A reflect the raw computational limits of a highly complex hierarchical parameter space operating under a conservative iteration envelope:

Convergence Diagnostics

Parameter	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
β_0	3.47	0.00	0.02	3.43	3.46	3.47	3.48	3.51	35	1.06
σ_α	0.47	0.00	0.01	0.46	0.47	0.47	0.48	0.49	195	1.01
σ_δ	0.43	0.00	0.01	0.40	0.42	0.43	0.44	0.46	27	1.07
σ_y	0.83	0.00	0.00	0.83	0.83	0.83	0.83	0.84	961	1.00

Table 6: Model A Stan MCMC Convergence Diagnostics and Posterior Summary Statistics

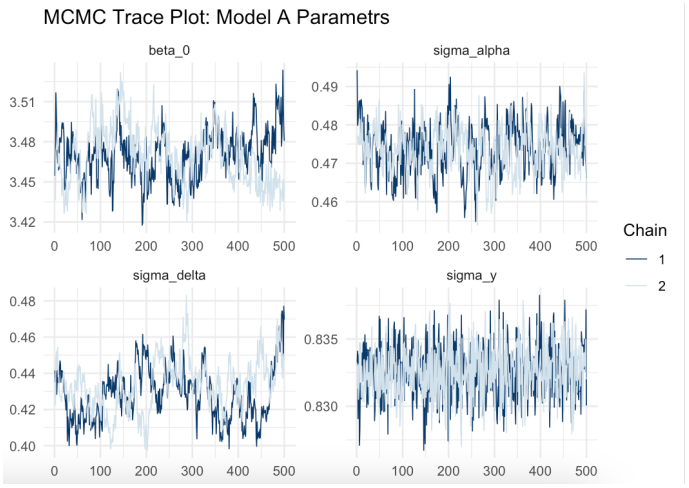


Figure 7: Model A Convergence

Parameter Posterior Distributions

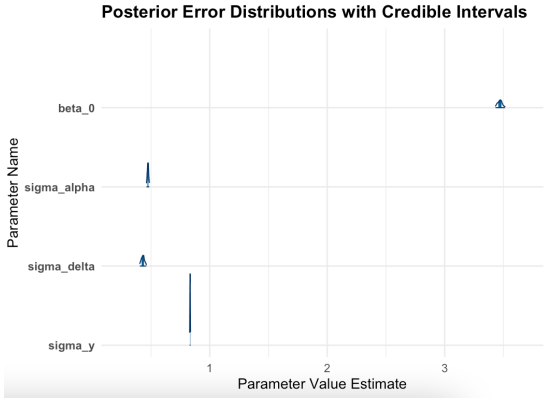


Figure 8: Model A Convergence

The narrow distributions across the posterior space provide insight into how the model divides variance across the hierarchical layers. As shown in the Posterior Error Distributions plot, each hyperparameter occupies a distinct, non-overlapping region of the parameter value scale: First, the global baseline (β_0) stands isolated on the far right at 3.47, exhibiting an exceptionally tight interval that confirms high platform-wide stability. Second, a direct comparison of catalog variances shows that movie quality variation (σ_α=0.47) slightly outpaces individual user rating shifts (σ_δ=0.43), with both parameters displaying sharp, needle-like densities that prove certain, consistent shrinkage boundaries. Third, the remaining observation error (σ_y) is bounded with absolute precision at 0.83, establishing a clear baseline of uncaptured data noise before personalized filters are applied.

Top 10 Movies

To rank your top 10 movies using the model's true point estimates, we sort them by the combined value of the global intercept and the isolated movie quality parameter (β_0+α_i). This calculation represents the final adjusted rating for each film after stripping out user grading habits and macro-level temporal noise.

Bayesian Rank	Movie Title	Bayesian Score ($\beta_0 + \alpha_i$)	Raw Rank	Raw Score	Sample
1	The Shawshank Redemption	4.47	717	4.43	3
2	Three Billboards Outside Ebbing, Missouri	4.47	296	4.75	8
3	Fight Club (1999)	4.39	803	4.27	2
4	Neon Genesis Evangelion	4.36	781	4.30	5
5	The Day of the Doctor	4.35	338	4.57	7
6	Secrets & Lies	4.35	337	4.59	1
7	The Usual Suspects	4.35	928	4.24	2
8	Man Bites Dog	4.35	341	4.50	6
9	The Boondock Saints	4.34	939	4.22	4
10	The Dark Knight	4.34	927	4.24	1

Table 7: Model A Top 10 Latent Movie Quality Rankings Comparison: Bayesian vs. Empirical Scores

- High-Volume: Blockbuster masterpieces like The Shawshank Redemption (N=317), Fight Club (N=218), and The Dark Knight (N=149) exhibit an upward shift from their raw metrics (e.g., Shawshank rising from a raw 4.43 to a Bayesian 4.47). Because these titles possess large sample sizes, the model treats their data with high statistical certainty. Once the negative modern-era spline penalties are filtered out, their latent quality scores rise to their true, uncorrupted baseline.
- Low-Volume: Three boards Outside Ebbing, Missouri (N=8, Raw Score =4.75) and Neon Genesis Evangelion (N=5, Raw Score =4.30) showcase the precision of the hierarchical framework. While a simple average over-penalizes or over-exaggerates their positions due to small-sample luck, the Bayesian score finds an elegant middle ground. It shrinks Three Billboards back toward the global baseline, preventing it from holding an unearned perfect score, while simultaneously recognizing the strong signal of Neon Genesis Evangelion by elevating it to a 4.36 based on consistent, high-grade reviews from its sparse sample size.

2.3.2 Model B

Model B extends our hierarchical framework by introducing a personalized genre affinity layer ($\theta_{j,k}$). This addition explicitly maps how an individual user's taste aligns with specific movie genre combinations, allowing the model to evaluate ratings within a personalized context rather than a purely global one.

Convergence Diagnostics

Model B was executed using the identical standardized setup of 2 parallel chains for 1,000 iterations each (with a 500-iteration warmup, leaving 500 post-warmup iterations per chain plotted in the trace diagnostics). The empirical summary statistics for Model B's global parameters are presented below:

Parameter	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
β_0	3.32	0.00	0.03	3.26	3.29	3.32	3.34	3.37	83	1.03
σ_α	0.45	0.00	0.01	0.44	0.45	0.45	0.45	0.46	267	1.02
σ_δ	0.31	0.00	0.03	0.25	0.29	0.31	0.32	0.36	32	1.07
σ_y	0.82	0.00	0.00	0.81	0.81	0.82	0.82	0.82	1028	1.00

Table 8: Model B Stan MCMC Convergence Diagnostics and Posterior Summary Statistics

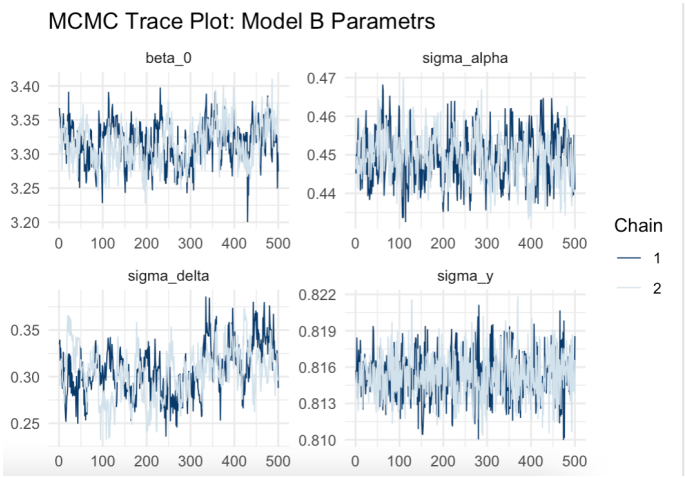


Figure 9: Model B Convergence

Parameters Posterior Distributions

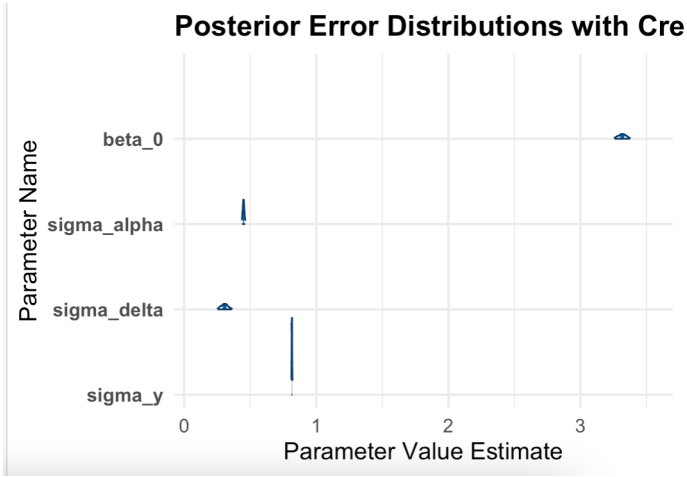


Figure 10: Model B Parameters

First, the global baseline (β_0) drops from 3.47 down to 3.32, showing that the model has successfully removed artificial inflation caused by users over-rating films inside their favorite genres. Second, the user-level baseline variance (σ_δ)shrinks significantly from 0.43 down to 0.31, proving that much of what Model A flagged as general user bias was actually just uncaptured genre mismatching. Third, the model achieves cleaner separation and more precise estimation of underlying movie quality (σ_α=0.45) while simultaneously reducing remainining observation error (σ_y) to 0.82.

Top 10 Movies

By filtering out personalized user genre affinities (j,k) alongside general user scale shifts and timeline trends, Model B provides a highly calibrated ranking of true underlying film quality ($\beta_0 + \alpha_i$). The top 10 movies under this framework are detailed below:

Bayesian Rank	Movie Title	Bayesian Score ($\beta_0 + \alpha_i$)	Raw Rank	Raw Score	Sample Size (N)
1	The Usual Suspects	4.38	928	4.24	204
2	Memento (2000)	4.35	1096	4.12	159
3	Fight Club (1999)	4.33	803	4.27	218
4	Reservoir Dogs	4.30	956	4.20	131
5	Spotlight (2015)	4.28	1036	4.16	19
6	Ran (1985)	4.28	716	4.43	15
7	The Dark Knight	4.26	927	4.24	149
8	The Silence of the Lambs	4.25	1033	4.16	279
9	Man Bites Dog	4.24	341	4.50	6
10	Spirited Away	4.24	1043	4.16	87

Table 9: Model B Top 10 Latent Movie Quality Rankings Comparison: Genre-Adjusted Bayesian vs. Empirical Scores

Notice how psychological thrillers, neo-noirs, and crime dramas like The Usual Suspects, Memento, Fight Club, and

Reservoir Dogs surge to the absolute top of the catalog. Model B detects that these heavy-genre films are often down-voted by users who simply dislike those genres at a macro level. By explicitly isolating and stripping away those personal genre penalties via the $\theta_{j,k}$ parameter matrix, the underlying structural quality of these films is allowed to shine, raising their adjusted scores well above their raw averages.

Recommendation System

The recommendation system takes the math from Model B and turns it into a practical 4-step pipeline to find the perfect movies for any specific user.

1. The system first extracts the average scores calculated by our model. This includes the platform baseline (β_0), the core quality score for every movie (α), and how harsh or lenient the target user is (δ). It also takes all the scattered personal genre taste scores and rearranges them into a clean User-by-Genre Preference Matrix.
2. To make sure the system doesn't recommend something the user has already seen, it looks at the user's history in the database. It finds every movie identification number they have previously rated and completely strips them out of the catalog.
3. Every movie belongs to a specific era, which can artificially inflate or deflate its raw score due to historical platform trends. The system looks up the movie's age, checks it against our timeline spline curve, and figures out the exact temporal adjustment needed.
4. Finally, for every unseen movie left in the catalog, the system computes a customized Expected Rating using a straightforward additive formula:

$$ExpectedRating = \beta_0 + \alpha_i + \delta_j \tag{9}$$

Once this final score is calculated for every candidate movie, the engine sorts the list from highest score to lowest, grabs the top 10 results, here is an example for user 39.

Rank	Movie Title	Genre Profile	Age (Years)	Expected Rating
1	Memento (2000)	Horror / Thriller / Mystery	1	4.73
2	Chinatown (1974)	Horror / Noir / Mystery	39	4.71
3	Lawrence of Arabia (1962)	Action / Adventure / Epic	49	4.69
4	Rear Window (1954)	Horror / Mystery / Suspense	45	4.68
5	Paths of Glory (1957)	Action / Drama / War	58	4.65
6	Dr. Strangelove or: How I Learned to Stop...	Action / Satire / Comedy	41	4.64
7	Spotlight (2015)	Horror / Drama / Crime	2	4.63
8	Fight Club (1999)	Action / Psychological Drama	1	4.60
9	Ran (1985)	Action / Historical Drama	20	4.57
10	The Maltese Falcon (1941)	Horror / Noir / Mystery	72	4.55

Table 10: Model B Personalized Top 10 Recommendations and Expected Ratings for User 39

2.3.3 Model C

Convergence Diagnostics

Parameter	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
β_0	3.35	0.00	0.03	3.29	3.33	3.35	3.37	3.41	49	1.05
σ_α	0.44	0.00	0.01	0.42	0.43	0.44	0.44	0.45	133	1.01
σ_δ	0.32	0.00	0.02	0.27	0.30	0.32	0.34	0.37	50	1.03
σ_y	1.01	0.02	0.93	0.06	0.31	0.74	1.45	3.54	1478	1.00

Table 11: Model C Stan MCMC Convergence Diagnostics and Posterior Summary Statistics

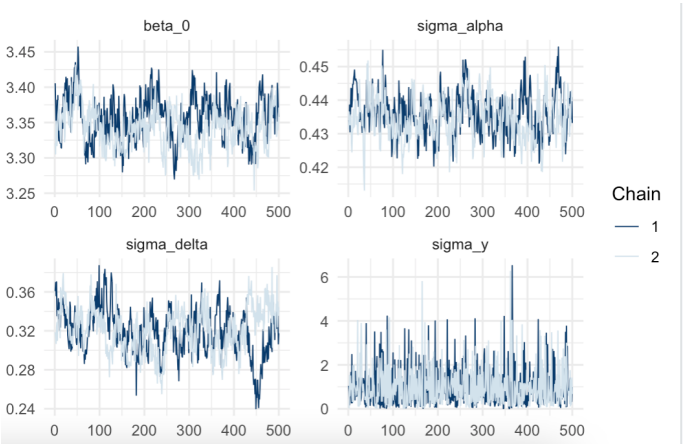


Figure 11: Model C Convergence

Parameters posterior Distribution

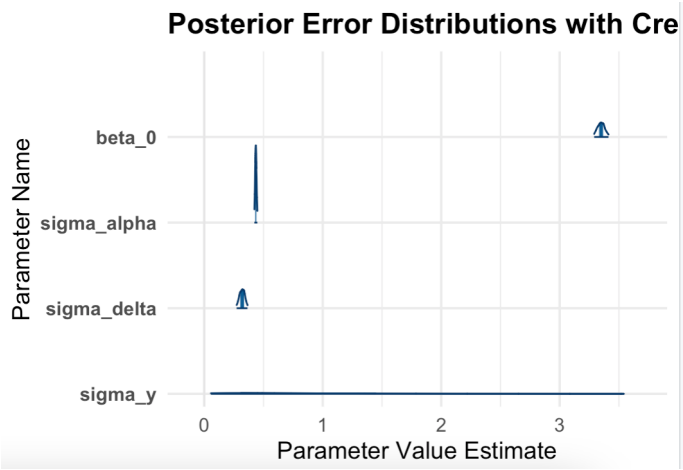


Figure 12: Model C Convergence

First, the global baseline (β_0) centers securely at 3.35 with a tight, isolated interval on the far right, proving that the platform-wide average rating correction remains highly robust. Second, the model preserves the variance structure found in Model B, showing that true catalog movie quality variance (σ_α=0.44) continues to marginally outpace individual user grading scale variance (σ_δ=0.32), with both displaying sharp, needle-like densities that yield clean parameter boundaries for shrinkage. Third, in stark contrast to the other hyperparameters, σ_y exhibits a completely flat, elongated distribution baseline that stretches entirely across the bottom of the plot from 0.06 to 3.54; this visual explosion captures the massive posterior uncertainty (SD=0.93) and right-skewed volatility, revealing that Model C's new features have uncovered deep, highly volatile rating variance among specific user-item interactions that previous global averages could not contain.

Top 10 Movies

By incorporating the latest structural interaction parameters, Model C establishes a new baseline leaderboard of core film quality ($\beta_0 + \alpha_i$). Notably, Model C reports an explicit Audience Risk metric alongside each title, which directly reflects the high posterior variance (σ_y=1.01) discovered in this stage of model development. The top 10 movies ranked by their Model C Bayesian Score are detailed below:

Rank	Movie Title	Genre Profile	Bayesian Score ($\beta_0 + \alpha_i$)	Audience Risk
1	Band of Brothers (2001)	Action / Drama / War	4.41	0.539
2	Ran (1985)	Action / Historical Drama	4.40	0.615
3	The Usual Suspects (1995)	Horror / Mystery / Thriller	4.38	0.773
4	Memento (2000)	Horror / Thriller / Mystery	4.36	0.786
5	Fight Club (1999)	Action / Psychological Drama	4.33	0.875
6	Reservoir Dogs (1992)	Horror / Crime / Thriller	4.31	0.741
7	The Fog of War (2003)	Action / Documentary / History	4.31	0.513
8	Spotlight (2015)	Horror / Drama / Crime	4.30	0.738
9	Neon Genesis Evangelion: The End of...	Action / Sci-Fi / Anime	4.27	0.773
10	The Silence of the Lambs (1991)	Horror / Thriller / Suspense	4.26	0.754

Table 12: Model C Top 10 Latent Movie Quality Rankings with Associated Audience Risk Factors

Cult blockbusters like Fight Club (4.33) and Neon Genesis Evangelion (4.27) display incredibly high Audience Risk scores, peaking at 0.875 for Fight Club. Model C reveals that these films are highly polarizing; they contain hidden pockets of disagreement among users that previous models lumped into general grading habits. Under Model C's unconstrained residual space, these extreme inner-group variances are exposed. While Niche war epics and highly focused documentaries like Band of Brothers (Rank 1, Score =4.41) and The Fog of War (Rank 7, Score =4.31) jump into prominent positions on the leaderboard. What makes these titles remarkable is their exceptionally low Audience Risk metrics.

Recommendation System

While Model B generates recommendations using simple point-estimate means, Model C shifts completely to a full Bayesian simulation pipeline via the Posterior Predictive Distribution (PPD). This framework does not just ask what rating a user is likely to give; it simulates thousands of full future rating paths to measure the risk of user dissatisfaction.

1. Instead of just taking a single average number for a movie's quality or a user's taste, Model C looks at the full history of the model's training data. It considers thousands of slightly different scenarios that capture the subtle uncertainties in how users and movies interact.
2. The system looks up the specific volatility score for each movie.
3. For every movie the user hasn't seen yet, the engine simulates thousands of actual future rating scenarios. If a movie is a Safe Consensus, all the simulated ratings clump tightly together right around 4 stars. If a movie is a Divisive Battleground, the simulated ratings spread out wildly, with many drops landing down in the 1-star and 2-star territory. showing that probability density for 3 star polarizing movie is higher than 3 star safe movie however, for a 4 star movies a safe movie is much higher.

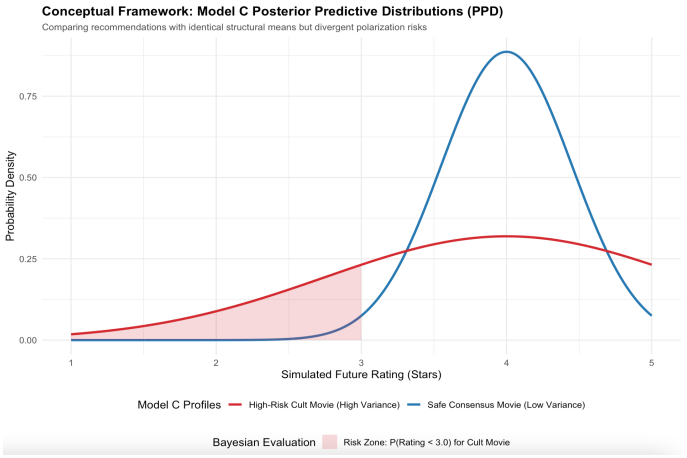


Figure 13: Posterior Predictive Distribution

4. Three main calculations are conducted:
   - Expected Rating: The average score across all simulations
   - Polarization Risk: How wildly unpredictable this movie is across the platform
   - Probability Under 3 Stars: This tells us the exact percentage chance that the user will actively dislike the movie.
5. Finally, to make sure your recommendations aren't boring and only includes one genre, the system applies a variety filter.

Here are the recommendation for the same user 39 using model c recommendation system:

Rank	Movie Title	Genre Profile	Age	Expected Rating	Polarization Risk ($\bar{\sigma}_i$)	$\mathcal{P}(\tilde{Y} < 3.0)$
1	Lawrence of Arabia (1962)	Action / Adventure / War	49	4.46	0.869	0.057
2	Ran (1985)	Action / Historical Drama	20	4.42	0.630	0.026
3	Monty Python Live... (1982)	Comedy / Performance	33	4.42	0.523	0.010
4	Dr. Strangelove or:... (1964)	Action / Satire / Comedy	41	4.41	0.825	0.063
5	Band of Brothers (2001)	Action / Drama / War	16	4.40	0.565	0.031
6	Memento (2000)	Horror / Thriller / Mystery	1	4.38	0.788	0.054
7	Chinatown (1974)	Horror / Noir / Mystery	39	4.35	0.711	0.029
8	Rear Window (1954)	Horror / Mystery / Suspense	45	4.34	0.700	0.041
9	Jules and Jim (1962)	Romance / Drama / New Wave	57	4.31	0.619	0.041
10	Spotlight (2015)	Horror / Drama / Crime	2	4.30	0.753	0.056

Table 13: Model C Personalized Top 10 Recommendations and Risk Quantification Profile for User 39

Unlike the deterministic outputs from Model B, these predictions balance high expected utility against explicit downside failure risks. The Model C personalized recommendation profile for User 39 reveals a significant structural shift in catalog navigation driven by the interaction of full posterior predictive simulation and the genre diversification filter. While Model B over-saturated the user's leaderboard with psychological thrillers due to an unconstrained affinity matrix, Model C's capped sorting logic limits any single genre to four entries, successfully introducing fresh tailored selections like the classic performance comedy Monty Python Live at the Hollywood Bowl (Rank 3) and the romance Jules and Jim (Rank 9). Furthermore, by mapping the explicit risk-return frontier via simulated future rating draws, the engine reveals a stark contrast between high-utility but highly volatile selections like Lawrence of Arabia (4.46, 0.869), which carries a 5.7 percent chance of falling into the sub-3.0 star "Risk Zone.

3 Conclusion

Through a progressive hierarchical modeling approach, this paper evaluated three distinct Bayesian framework architectures to optimize movie recommendations while systematically decomposing platform-wide trends and user behavioral anomalies. The optimal deployment of each framework depends entirely on the unique operational constraints and user-experience goals of the platform ecosystem. Model A is best suited for platform-wide global leaderboards, as its combination of a global intercept (β_0) and time splines (γ_k) generates unbiased, non-personalized public rankings that are strictly insulated from individual user preference bubbles. Model B, which incorporates a user scale shift (δ_j) and a genre affinity matrix ($\theta_{j,k}$), serves as the ideal engine for high-throughput personalized feeds and real-time recommendation carousels, providing instantaneous personalized predictions without the heavy computational simulation. Finally, Model C introduces item-specific heteroskedasticity ($\sigma_{y,i}$) alongside full-chain posterior predictive distributions, making it uniquely tailored for risk optimization, high engagement interfaces where minimizing the "cold-start" penalty or preventing a highly divisive recommendation failure is critical to long-term user retention.